

1) Setup Handout

Nextflow Local Installation Guide (Mac + Windows)

Goal

By the end of setup, you should be able to run:

```
java -version
nextflow -v
docker version
docker run --rm hello-world
nextflow run hello -with-docker
```

A. What you need

1. **Java 17 or newer** (Java runtime required by Nextflow)
2. **Nextflow** (workflow engine)
3. **Docker Desktop** (for reproducible containerized runs)
4. **Git** (recommended for pulling pipelines)

B. Mac setup

1) Install Homebrew (if missing)

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

2) Install Java + Git

```
brew install openjdk@21 git
```

3) Configure Java

```
echo 'export
JAVA_HOME="/opt/homebrew/opt/openjdk@21/libexec/openjdk.jdk/Contents/Home"'
>> ~/.zshrc
echo 'export PATH="$JAVA_HOME/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
java -version
```

4) Install Nextflow

```
curl -s https://get.nextflow.io | bash
sudo mv nextflow /usr/local/bin/
nextflow -v
```

5) Install Docker Desktop (Mac)

Install and launch Docker Desktop.

6) Validate Docker

```
docker version
docker run --rm hello-world
```

C. Windows setup (required method: WSL2 + Ubuntu)

1) Install WSL2

Open PowerShell as Administrator:

```
wsl --install
wsl --update
```

Reboot if prompted, then open Ubuntu and create your Linux username/password.

2) In Ubuntu terminal, install Java + Git + curl

```
sudo apt update
sudo apt install -y openjdk-21-jdk git curl
java -version
```

3) Install Nextflow (inside Ubuntu)

```
curl -s https://get.nextflow.io | bash
sudo mv nextflow /usr/local/bin/
nextflow -v
```

4) Install Docker Desktop (Windows)

Install Docker Desktop and enable WSL integration for your Ubuntu distro.

5) Validate inside Ubuntu

```
docker version
docker run --rm hello-world
```

D. First test pipeline

Run:

```
nextflow run hello
nextflow run hello -with-docker
```

If second command fails, Docker is usually not running yet.

E. Submit this

Take screenshots of:

```
java -version
nextflow -v
docker version
docker run --rm hello-world
```

2) Troubleshooting

Use this in order. Do not skip steps.

Step 1 Check Java

Run:

```
java -version
```

- If command not found or version <17: install/fix Java first.

Step 2 Check Nextflow

Run:

```
nextflow -v
```

- If command not found reinstall via `get.nextflow.io` and confirm PATH.

Step 3 Check Docker CLI

Run:

```
docker --version
```

- If command not found:
 - Docker Desktop not installed, or
 - CLI path missing from shell PATH.

Step 4 Check Docker daemon

Run:

```
docker version
```

- If it cannot connect to daemon:
 - Start Docker Desktop app
 - Wait until fully initialized
 - Retry `docker version`

Step 5 Check container run

Run:

```
docker run --rm hello-world
```

- If fails: Docker engine/context issue; restart Docker Desktop and retry.

Step 6 Check Nextflow + Docker integration

Run:

```
nextflow run hello -with-docker
```

- If error says “**no image was specified**”:
 - Ensure pipeline has `container 'image:tag'` per process

- OR set a default in `nextflow.config`:
- `process.container = 'ubuntu:22.04'`
- `docker.enabled = true`

Step 7 Resume after fixing

If a run failed and you fixed issue:

```
nextflow run <pipeline> -with-docker -resume
```

Step 8 Inspect task-level logs

For failed processes:

- Open the task work directory shown in error
- Inspect:
 - `.command.sh`
 - `.command.out`
 - `.command.err`

Step 9 Windows-specific reminders

- Must run in **Ubuntu WSL**, not PowerShell/CMD.
- Keep project files in Linux home (`~/project`) for best performance.

Official sources

- Nextflow installation/get started:

https://www.nextflow.io/docs/latest/install.html?utm_source=chatgpt.com

- Nextflow site/prereqs overview:

https://www.nextflow.io/?utm_source=chatgpt.com

- Nextflow containers docs:

https://nextflow.io/docs/latest/container.html?utm_source=chatgpt.com

- Docker Desktop (Mac):

https://docs.docker.com/desktop/setup/install/mac-install/?utm_source=chatgpt.com

- Docker Desktop (Windows):

https://docs.docker.com/desktop/setup/install/windows-install/?utm_source=chatgpt.com

- Docker Desktop overview/download hub:

https://docs.docker.com/get-started/introduction/get-docker-desktop/?utm_source=chatgpt.com

- Microsoft WSL install:

https://learn.microsoft.com/en-us/windows/wsl/install?utm_source=chatgpt.com